

9. Overview and concepts

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Cross-validation

Cross-validation is a resampling procedure for gaining the average estimate of a model's performance. Its objective is to test a model's ability to make predictions on a portion of data that is different from the dataset it was initially trained on.

Cross-validation applies to both regression and classification tasks. The simplest form of cross-validation is the % split between training and test sets (i.e. *one-round* cross-validation). Although we are familiar with this simple cross-validation procedure, there are reasons why it may be insufficient to evaluate a model's performance reliably. The simple % split is known to be prone to the *variability problem*, where a model's performance on the test set is determined more by how the data is split than by the inherent quality of the model. This is evident from the changes we usually observe in a model's performance scores responding to changes in the % or the random seed value.

k-fold cross-validation

To overcome the limitations of the one-round cross-validation, it is recommended to use *k*-fold cross-validation. This technique randomly splits a dataset into *k* equally-sized segments, of which *k*-1 segments are used for training and the remaining segment is used for testing.

The following animation illustrates k-fold cross-validation for $k=3$.

$n = 12$
 $k = 3$

 Test  Train

Data



Source: wikipedia.com_([https://en.wikipedia.org/wiki/Cross-validation_\(statistics\)#k-fold_cross-validation](https://en.wikipedia.org/wiki/Cross-validation_(statistics)#k-fold_cross-validation))

The stepwise procedure is as follows:

- 1) Shuffle records in the dataset randomly.
- 2) Split the dataset into k segments.
- 3) For each segment:
 - Take the segment as a test data set
 - Take the remaining segments as a training data set
 - Fit a model on the training set and evaluate it on the test set
 - Retain the evaluation score and discard the model
- 4) Iterate Step 3 k number of times until all segments are used as a test set. Summarize the average performance of the model over the iterations.

The common settings for k are $k=5$ and $k=10$.

Source: machinelearningmastery.com_(<https://machinelearningmastery.com/k-fold-cross-validation/>)

Confusion Matrix

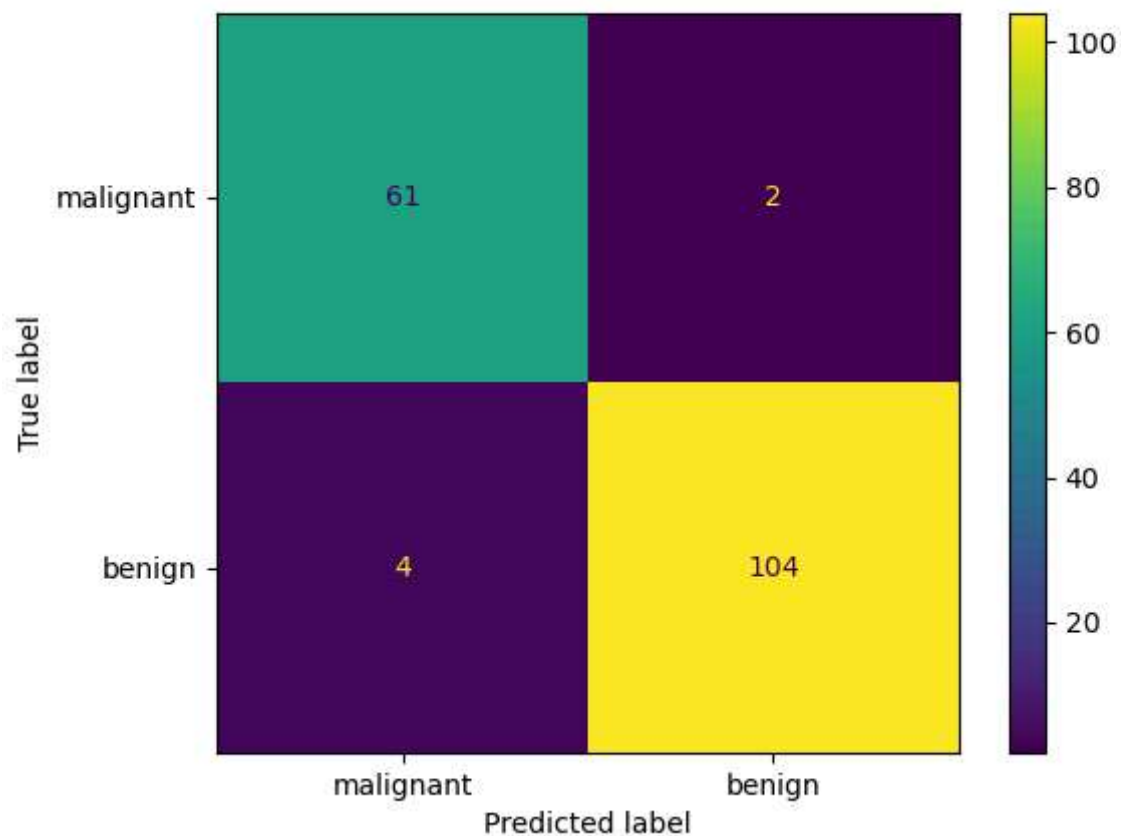
The confusion matrix visually depicts the performance of a **classifier** (not for regression) in terms of the number of actual versus predicted class labels. Typically, matrix rows represent the actual class labels, and columns represent the predicted labels. Numbers in the intersections between these rows and columns (i.e. cells) signify the amount of accurate and inaccurate classifications.

Here is the anatomy of the confusion matrix:

		Predicted condition	
Total population = P + N		Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP)	False negative (FN)
	Negative (N)	False positive (FP)	True negative (TN)

Source: wikipedia.com_(https://en.wikipedia.org/wiki/Confusion_matrix).

Below is an example confusion matrix for classifying malignant and benign breast cancer tissues.



Several important classification metrics can be calculated from a confusion matrix:

- Accuracy
- Recall
- Precision
- F1

Look up the equations of these metrics: https://en.wikipedia.org/wiki/Confusion_matrix
(https://en.wikipedia.org/wiki/Confusion_matrix).

The following video further explains the confusion matrix.

Machine Learning Fundamentals: The Confusion Matrix



Receiver Operating Characteristic (ROC) curve

This video explains the intuition behind the ROC curve.

ROC and AUC, Clearly Explained!

